Finding the Centre & Radius

Exam code: 9MA0

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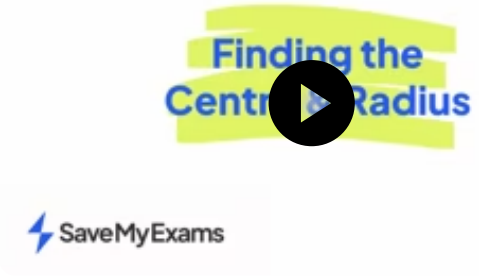
Updated on 20 June 2025



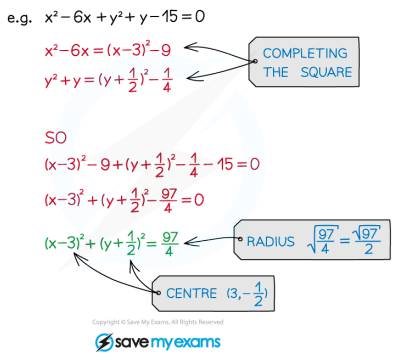
Did this video help you? Yes No

Help me understand

Finding the centre & radius

How do I find the centre and radius from the equation of a circle?

- A circle equation in a different form can always be rearranged into $(x-a)^2 + (y-b)^2 = r^2$ form in order to find the centre and radius
- This will often involve **completing the square**



Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

the general form $x^2 + y^2 + 2fx + 2gy + c = 0$

$$x^2 + y^2 + 2fx + 2gy + c = 0$$
$$x^2 + 2fx + y^2 + 2gy + c = 0$$
$$(x+f)^2 - f^2 + (y+g)^2 - g^2 + c = 0$$
$$(x+f)^2 + (y+g)^2 = f^2 + g^2 - c$$

RADIUS $\sqrt{f^2+g^2-c}$

CENTRE $(-f, -g)$

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Worked Example

A circle has the equation

$$x^2 + 2x + y^2 - 6y + k = 0$$

and passes through the point $(5, 12)$.

Find the centre and radius of the circle.

CIRCLE PASSES THROUGH (5,12) SO

$$(5)^2 + 2(5) + (12)^2 - 6(12) + k = 0$$
$$25 + 10 + 144 - 72 + k = 0$$
$$k = -107$$

USE THE COORDINATES OF THE KNOWN POINT TO FIND THE VALUE OF k

EQUATION OF THE CIRCLE IS

$$x^2 + 2x + y^2 - 6y - 107 = 0$$
$$(x+1)^2 - 1 + (y-3)^2 - 9 - 107 = 0$$
$$(x+1)^2 + (y-3)^2 = 117$$

REARRANGE INTO $(x-a)^2 + (y-b)^2 = r^2$ FORM

CENTRE IS $(-1, 3)$

RADIUS IS $\sqrt{117}$

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